Instructions. Please write up your solutions on a separate sheet of paper.

Problem 1

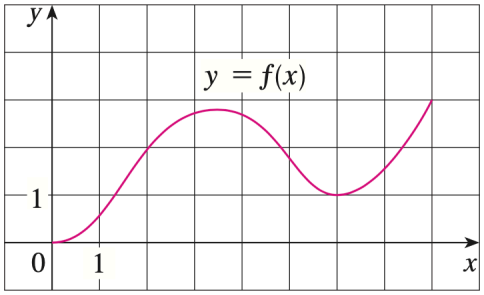
Consider the function $f(x) = \sqrt{x} - \frac{1}{3}x$ defined on $[0,9]$. Show that f satisfies the hypotheses of Rolle's Theorem. Find all numbers c that satisfy the conclusion of Rolle's Theorem.

Problem 2

Let $f(x) = 1 - x^{2/3}$. Show that $f(-1) = f(1)$ but there is no number c in $(-1,1)$ such that $f'(c) = 0$. Why does this not contradict Rolle's Theorem?

Problem 3

Use the graph of f to estimate the values of c that satisfy the conclusion of the Mean Value Theorem for the interval $[1,7]$.



Problem 4

Consider the function $f(x) = \frac{1}{x}$ on the interval $[1,3]$. Verify that f satisfies the hypotheses of the Mean Value Theorem on the given interval. Find all numbers c that satisfy the conclusion of the Mean Value Theorem.

Problem 5

Let $f(x) = (x-3)^{-2}$. Show that there is no value of c in $(1,4)$ such that $f'(c) = \frac{f(4)-f(1)}{(4-1)}$. Why does this not contradict the Mean Value Theorem?

Problem 6

Let f be a differentiable function defined on an interval. Show that if $f'(x) = c$ (where c is a constant) for all x, then $f(x) = cx + d$ for some constant d.

Problem 7

Suppose that f and g are continuous on $[a,b]$ and differentiable on (a,b). Suppose also that $f(a) = g(a)$ and $f'(x) < g'(x)$ for $a < x < b$. Prove that $f(b) < g(b)$. *Hint: Apply the Mean Value Theorem to the function $h = f - g$.*

Problem 8

Show that the equation $x^3 - 15x + c = 0$ (where c is a constant) has at most one root in the interval $[-2,2]$.

Problem 9

If $f(1) = 10$ and $f'(x) \geq 2$ for $1 \leq x \leq 4$, how small can $f(4)$ possibly be?

Problem 10

Does there exist a function f such that $f(0) = -1$, $f(2) = 4$, and $f'(x) \leq 2$ for all x?

Problem 11

Show that $\sqrt{1+x} < 1 + \frac{1}{2}x$ if $x > 0$. *Hint: Use the result in Problem 8.*

Problem 12

Suppose f is an odd function and is differentiable everywhere. Prove that for every positive number b, there exists a number c in $(-b,b)$ such that $f'(c) = f(b)/b$.

Problem 13

A number a is called a *fixed point* of a function f if $f(a) = a$. Suppose that f is differentiable for all x. Prove that if $f'(x) \neq 1$ for all real numbers x, then f has at most one fixed point.

Problem 14 (Optional)

Suppose g has a bounded derivative (i.e., there exists a positive number M such that $|g'(x)| < M$ for all x). Fix $\varepsilon > 0$, and define $f(x) = x + \varepsilon g(x)$. Prove that f is one-to-one if ε is small enough, and determine the largest value of ε for which f is one-to-one. Give a concrete example of f by choosing an appropriate function g, and give the largest value of ε for which f is one-to-one. Use Desmos to plot the graph of f for several values of ε, and observe that f passes the horizontal line test if ε is less than or equal to the maximum value you determined, but fails when ε is larger. Include a screen shot with your solution.